

[Foundational Clinic 3] Grade 4-5 Fraction Basics, Equivalent Fractions & Operations

Student Name: _____

Target: _____

Time: 35

Total: _____

Score: _____

🌸 Sophia & ⚡ William · Equivalent Fractions, Mixed Numbers, Division Reciprocal

Mins

100 Pts

💡 Key Fraction Laws (Root Error Correction):

① **Equivalent Fractions:** Whatever you multiply/divide the denominator by, you MUST do the exact same to the numerator! $\frac{3}{4} = \frac{3 \times 5}{4 \times 5} = \frac{15}{20}$. Never just multiply the denominator and forget the numerator!

② **Division with Remainder to Mixed Number:** In $52 \div 24 = 2 \text{ R } 4$, the quotient 2 is the whole number, the remainder 4 is the numerator, and 24 remains the denominator: $2\frac{4}{24} = 2\frac{1}{6}$ (NEVER write $24\frac{4}{2}$!).

③ **Fraction Division:** Keep, Change, Flip! $a \div \frac{b}{c} = a \times \frac{c}{b}$. Always multiply by the reciprocal.

④ **Borrowing in Mixed Subtraction:** When $3\frac{1}{5} - 1\frac{4}{5}$, borrow 1 = $\frac{5}{5}$ from 3: $3\frac{1}{5} = 2\frac{6}{5}$. Then $2\frac{6}{5} - 1\frac{4}{5} = 1\frac{2}{5}$.

Section 1: Equivalent Fractions & Simplifying (3 pts each, 18 pts total)

Score: _____

Fill in the missing numerators or denominators to make the fractions equivalent, or simplify to lowest terms:

(a) $\frac{3}{7} = \frac{\quad}{28}$ _____

(b) $\frac{5}{8} = \frac{35}{\quad}$ _____

(c) $\frac{18}{24} = \frac{\quad}{4}$ _____

(d) Simplify $\frac{45}{60} =$ _____

(e) Simplify $\frac{28}{42} =$ _____

(f) $\frac{4}{9} = \frac{\quad}{36}$ _____

Section 2: Improper Fractions & Mixed Numbers Conversion (3 pts each, 18 pts total)

Score: _____

Express as mixed numbers in simplest form (Quotient $\frac{\text{Remainder}}{\text{Divisor}}$) or convert to improper fractions:

(a) $52 \div 24 =$ _____

(b) $\frac{38}{6} =$ _____

(c) $\frac{47}{5} =$ _____

(d) Convert $4\frac{3}{7}$ to improper fraction: _____

(e) Convert $5\frac{2}{9}$ to improper fraction: _____

(f) $75 \div 12 =$ _____

Section 3: Fraction Addition & Subtraction (Common Denominator) (4 pts each, 24 pts total)

Score: _____

Show your common denominator calculation. Do NOT simply add across numerators and denominators!

(a) $\frac{3}{4} + \frac{2}{5} =$
LCD = 20. Expand BOTH numerators! _____

(b) $\frac{5}{6} - \frac{3}{8} =$
LCD = 24. _____

(c) $2\frac{1}{3} + 1\frac{3}{5} =$
Add whole numbers and fractions. _____

(d) $3\frac{1}{4} - 1\frac{3}{4} =$
Borrow 1 from 3 first! _____

(e) $4\frac{2}{7} - 2\frac{5}{7} =$ _____

(f) $\frac{7}{10} - \frac{1}{4} + \frac{2}{5} =$ _____

$$4\frac{2}{7} = 3\frac{9}{7}$$

Find LCD of 10, 4, 5 (LCD = 20).

Section 4: Fraction Multiplication & Division (4 pts each, 20 pts total)

Score: _____

Simplify before multiplying where possible. For division, multiply by the reciprocal (Keep-Change-Flip):

(a) $\frac{4}{9} \times \frac{3}{8} =$ _____

(b) $\frac{5}{12} \times 16 =$ _____

(c) $\frac{7}{10} \div \frac{14}{15} =$ _____

(d) $6 \div \frac{3}{4} =$ _____

(e) $2\frac{1}{4} \times 1\frac{1}{3} \div \frac{9}{10} =$ (Convert to improper fractions first!) _____

Section 5: Comparing Fractions & Word Applications (5 pts each, 20 pts total) Score: _____1. [Comparing Fractions] Insert $>$, $<$, or $=$ in each blank:

(a) $\frac{5}{12}$ _____ $\frac{7}{16}$

(b) $\frac{11}{12}$ _____ $\frac{12}{13}$ (Residual method)

(c) $\frac{7}{9}$ _____ $\frac{7}{11}$ (Same numerator)

2. [SUIS Practical Problem - Recipe Partition]

A baking recipe requires $2\frac{1}{4}$ kg of flour. Sophia has a bag containing $\frac{3}{5}$ kg of flour and William brings another bag containing $1\frac{1}{2}$ kg of flour.

(a) What is the total mass of flour they have together?

(b) Do they have enough flour for the recipe? If not, how many kilograms of flour are they short of?

Working & Verification:

3. [Fraction of a Remaining Quantity]

A water tank has a capacity of 480 liters. On Monday, $\frac{3}{8}$ of the water was used. On Tuesday, $\frac{2}{5}$ of the **remaining** water was used. How many liters of water are left in the tank?**Step-by-step Solution:** **Complete Answer Key & Pedagogical Notes****Section 1 (Equivalent):**

- (a) $12 (3 \times 4)$; (b) $56 (8 \times 7)$; (c) $3 (18 \div 6)$;
(d) $\frac{3}{4}$ (divide by 15); (e) $\frac{2}{3}$ (divide by 14); (f) $16 (4 \times 4)$.

Section 3 (Add/Subtract):

- (a) $\frac{15}{20} + \frac{8}{20} = \frac{23}{20} = 1\frac{3}{20}$;
(b) $\frac{20}{24} - \frac{9}{24} = \frac{11}{24}$;
(c) $2\frac{5}{15} + 1\frac{9}{15} = 3\frac{14}{15}$;
(d) $2\frac{3}{4} - 1\frac{3}{4} = 1\frac{2}{4} = 1\frac{1}{2}$ (borrow 1 as $\frac{4}{4}$);
(e) $3\frac{9}{7} - 2\frac{5}{7} = 1\frac{4}{7}$;
(f) $\frac{14}{20} - \frac{5}{20} + \frac{8}{20} = \frac{17}{20}$.

Section 2 (Conversions):

- (a) $2\frac{4}{24} = 2\frac{1}{6}$ (Sophia & William check: do not flip quotient & remainder!);
(b) $6\frac{2}{6} = 6\frac{1}{3}$; (c) $9\frac{2}{5}$;
(d) $\frac{31}{7} (4 \times 7 + 3)$; (e) $\frac{47}{9}$; (f) $6\frac{3}{12} = 6\frac{1}{4}$.

Section 4 & 5 (Mult/Div & Applications):

- Sec 4: (a) $\frac{1}{6}$; (b) $\frac{20}{3} = 6\frac{2}{3}$; (c) $\frac{7}{10} \times \frac{15}{14} = \frac{3}{4}$; (d) $6 \times \frac{4}{3} = 8$;
(e) $\frac{9}{4} \times \frac{4}{3} \times \frac{10}{9} = \frac{10}{3} = 3\frac{1}{3}$.
Sec 5: 1. (a) $\frac{5}{12} = \frac{20}{48} < \frac{21}{48} = \frac{7}{16}$; (b) $1 - \frac{1}{12} < 1 - \frac{1}{13} \implies \frac{11}{12} < \frac{12}{13}$; (c) $\frac{7}{9} > \frac{7}{11}$.
2. (a) $\frac{3}{5} + 1\frac{1}{2} = \frac{6}{10} + 1\frac{5}{10} = 2\frac{11}{10}$ kg; (b) No; $2\frac{1}{4} - 2\frac{1}{10} = 2\frac{5}{20} - 2\frac{2}{20} = \frac{3}{20}$ kg short.
3. Remaining after Mon: $480 \times (1 - \frac{3}{8}) = 300$ L.

Tue used: $300 \times \frac{2}{5} = 120$ L. Left: $300 - 120 = 180$ L.